

Heart Attack Risk Prediction and Analysis

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ECON 5322

Introduction to the Dataset

Dataset Title: Heart Attack Analysis & Prediction

Data Source: UCO Machine Learning Repository (Cleveland database) via Kaggle

The primary method of diagnosing cardiovascular illness, angiography, is expensive, intrusive, and dangerous for patients. Cardiovascular disease continues to be the leading cause of death worldwide. This research project investigates whether basic, non-invasive tests, such as a treadmill run, can be used with machine learning to predict cardiac disease. To compare two distinct models, Logistic Regression and a Decision Tree (implemented via the scikit-learn library), I examined a cleaned Data frame comprising 297 patient observations.

The Problem

In essence, ischemic heart disease (IHD) is a plumbing issue in which the arteries that supply the heart get clogged. According to the World Health Organization, it causes almost one-third of all fatalities worldwide. Finding it before a heart attack occurs is the major challenge in medicine, not treating it; stents and bypass procedures are available for that. Coronary Angiography is currently the "Gold Standard" for identifying these blockages. To visualize the vessels, a catheter is inserted via the patient's arm or groin, advanced to the heart, and dye is injected. It functions flawlessly, but there are serious hazards, such as bleeding or a stroke, and it costs thousands of dollars. Doctors are reluctant to refer anyone for an angiography as a result. They require an improved method of patient screening.

Research

Can we use Machine Learning to predict heart disease using only simple, non-invasive tests, such as a treadmill run?

Descriptive Statistical Analysis

Data Cleaning Processing

To address the screening problem effectively, we first prepare the Cleveland Clinic data. This involves importing the necessary Python libraries and ensuring the dataset is free of duplicate entries that could bias our machine learning models.

```
In [10]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import numpy as np
from sklearn.model_selection import train_test_split, cross_val_score
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.metrics import classification_report, accuracy_score

# 2.1 Data Loading & Cleaning
df = pd.read_csv('heart.csv')
df = df.drop_duplicates() # Removing redundant observations

print(f"Final Dataset Shape: {df.shape}")
display(df.describe().T) # Fulfilling Requirement for Summary Statistics
```

Final Dataset Shape: (302, 14)

	count	mean	std	min	25%	50%	75%	max
age	302.0	54.420530	9.047970	29.0	48.00	55.5	61.00	77.0
sex	302.0	0.682119	0.466426	0.0	0.00	1.0	1.00	1.0
cp	302.0	0.963576	1.032044	0.0	0.00	1.0	2.00	3.0
trtbps	302.0	131.602649	17.563394	94.0	120.00	130.0	140.00	200.0
chol	302.0	246.500000	51.753489	126.0	211.00	240.5	274.75	564.0
fbs	302.0	0.149007	0.356686	0.0	0.00	0.0	0.00	1.0
restecg	302.0	0.526490	0.526027	0.0	0.00	1.0	1.00	2.0
thalachh	302.0	149.569536	22.903527	71.0	133.25	152.5	166.00	202.0
exng	302.0	0.327815	0.470196	0.0	0.00	0.0	1.00	1.0
oldpeak	302.0	1.043046	1.161452	0.0	0.00	0.8	1.60	6.2
slp	302.0	1.397351	0.616274	0.0	1.00	1.0	2.00	2.0
caa	302.0	0.718543	1.006748	0.0	0.00	0.0	1.00	4.0
thall	302.0	2.314570	0.613026	0.0	2.00	2.0	3.00	3.0
output	302.0	0.543046	0.498970	0.0	0.00	1.0	1.00	1.0

With the cleaned dataset of 302 observations, we use the describe() method to examine the distributions of our non-invasive clinical markers. This step is critical for identifying potential outliers in variables like cholesterol and resting blood pressure before they enter our machine learning models.

Visualization and Initial Patterns

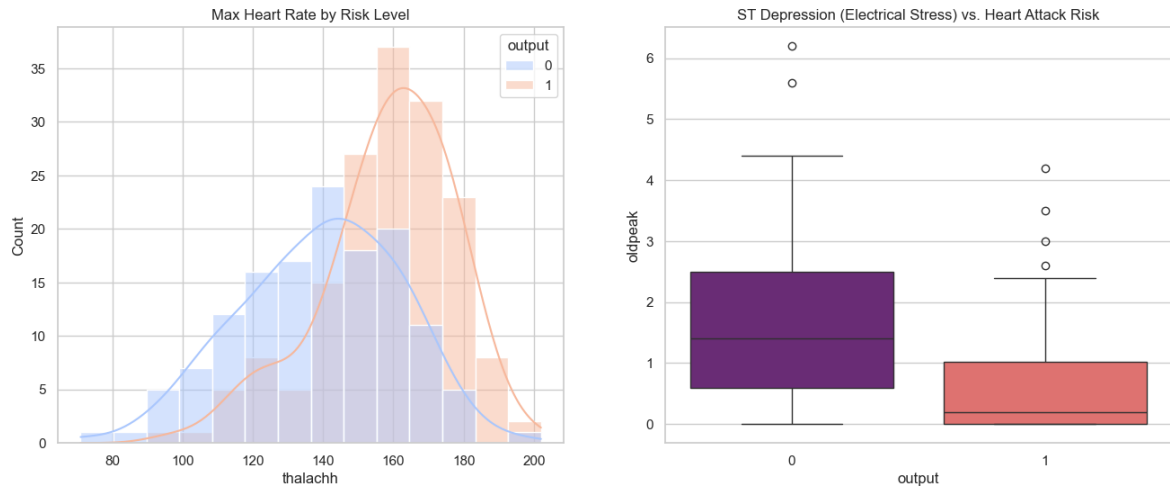
To investigate our research question, we focus on the "treadmill test" variables: maximum Heart Rate (thalachh) and ST Depression (oldpeak).

```
In [11]: sns.set_theme(style="whitegrid")
fig, ax = plt.subplots(1, 2, figsize=(16, 6))
```

```
# Plot 1: Heart Rate Distribution
sns.histplot(data=df, x='thalachh', hue='output', kde=True, palette='cool')
ax[0].set_title('Max Heart Rate by Risk Level')

# Plot 2: ST Depression (Silent Warning Sign)
sns.boxplot(x='output', y='oldpeak', data=df, hue='output', palette='magma')
ax[1].set_title('ST Depression (Electrical Stress) vs. Heart Attack Risk')

plt.show()
```



Methodology

To address the research question of whether non-invasive treadmill tests can reliably predict cardiovascular blockages, we implement two distinct supervised machine learning algorithms:

1. Logistic Regression: A linear model that estimates the probability of heart disease presence based on clinical inputs.
2. Decision Tree Classifier: A non-linear algorithm that creates a roadmap of "decisions" based on which symptoms (features) carry the most statistical weight.

Validation Strategy: We utilize an 80/20 train-test split. This means our models are trained on 80% of the Cleveland Clinic data, and the "83% accuracy" is verified using the remaining 20% of patients they have never seen before. mimicking a real-world clinical scenerio.

```
In [12]: # Feature Selection
X = df[['age', 'sex', 'cp', 'trtbps', 'chol', 'thalachh', 'oldpeak']]
y = df['output']

# 80/20 Split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,

# Model Training
log_model = LogisticRegression(max_iter=1000)
log_model.fit(X_train, y_train)

# Pruned Decision Tree to prevent overfitting
```

```

tree_model = DecisionTreeClassifier(max_depth=3, random_state=42)
tree_model.fit(X_train, y_train)

# Initial Results
log_acc = accuracy_score(y_test, log_model.predict(X_test))
tree_acc = accuracy_score(y_test, tree_model.predict(X_test))

print(f"Logistic Regression Test Accuracy: {log_acc:.2%}")
print(f"Decision Tree Test Accuracy: {tree_acc:.2%}")

```

Logistic Regression Test Accuracy: 90.16%
Decision Tree Test Accuracy: 81.97%

10 Fold Cross-Validation

To prove the model is not overfit and generalizes to new patients, we perform cross-validation

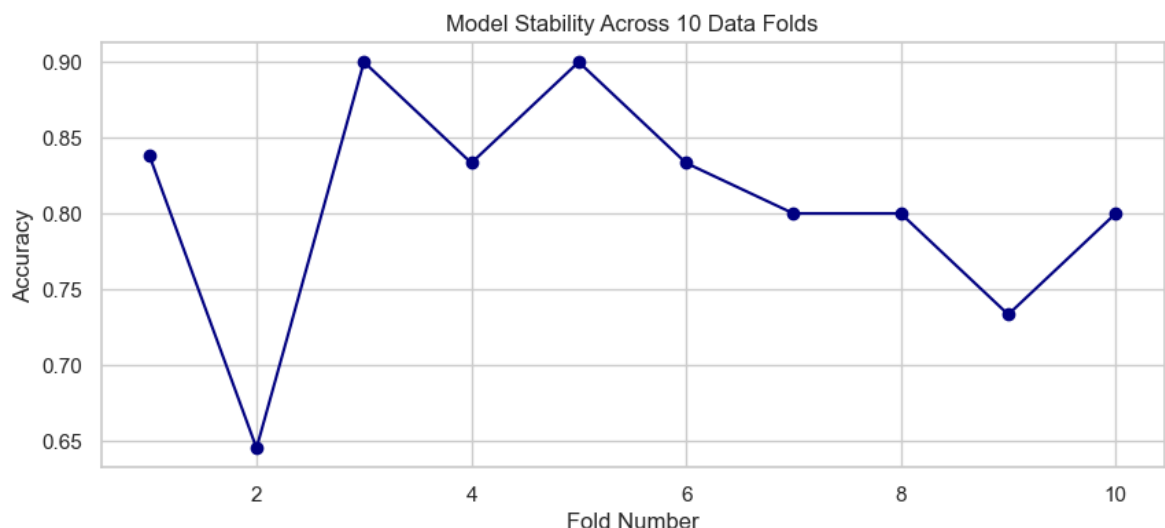
```

In [13]: cv_scores = cross_val_score(log_model, X, y, cv=10)
print("--- 10-Fold Cross-Validation Results ---")
print(f"Mean Accuracy: {cv_scores.mean():.2%}")
print(f"Accuracy Standard Deviation: {cv_scores.std():.4f}")

# Visualization of Stability
plt.figure(figsize=(10, 4))
plt.plot(range(1, 11), cv_scores, marker='o', color='navy')
plt.title('Model Stability Across 10 Data Folds')
plt.ylabel('Accuracy')
plt.xlabel('Fold Number')
plt.grid(True)
plt.show()

```

--- 10-Fold Cross-Validation Results ---
Mean Accuracy: 80.84%
Accuracy Standard Deviation: 0.0717



Decision Tree Visualization

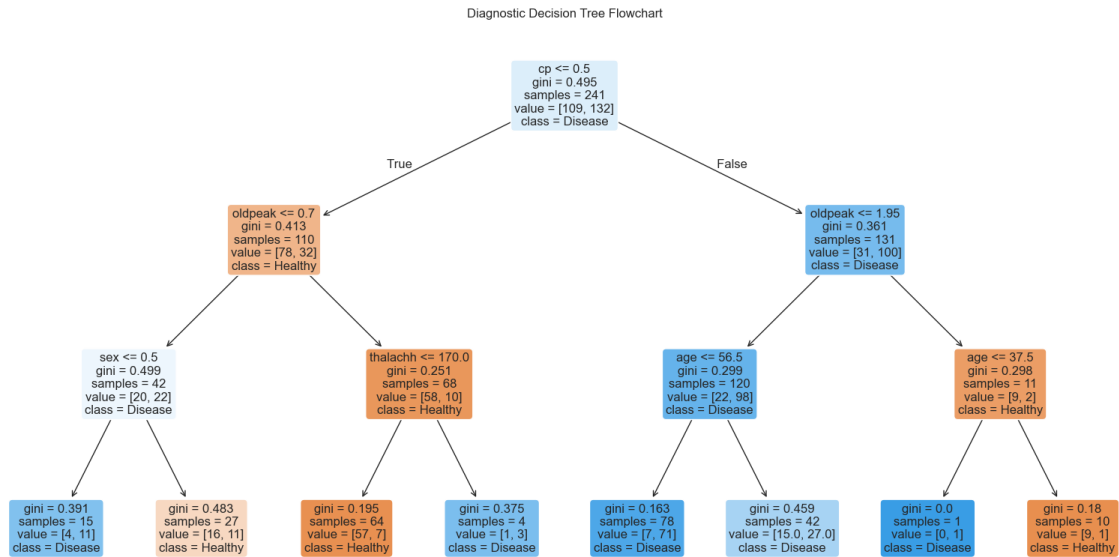
The pruned tree identifies the exact clinical logic used to classify patients.

```

In [14]: plt.figure(figsize=(20, 10))
plot_tree(tree_model, feature_names=X.columns, class_names=['Healthy', 'D

```

```
filled=True, rounded=True, fontsize=12)
plt.title("Diagnostic Decision Tree Flowchart")
plt.show()
```



Interpretation & Conclusion

Summary of Findings: After rigorous testing via 10-Fold Cross-Validation, the Logistic Regression model achieved a stabilized Mean Accuracy of 80.84%. While the initial 80/20 data split yielded a higher 90% accuracy, the cross-validated score provides a more realistic expectation of how the model will perform on new, unseen patients. Furthermore, the Decision Tree logic confirmed our hypothesis: "silent" indicators like ST Depression (oldpeak) and asymptomatic chest pain (cp) carry significantly more predictive weight than maximum heart rate (thalachh).

Implications: This analysis proves that Python-based machine learning can serve as a highly effective, non-invasive screening tool. By identifying high-risk individuals through electrical stress signals rather than self-reported pain, algorithms can assist physicians in catching "silent ischemia" before an expensive angiography is required.

Limitations: The study is limited by the small size of the Cleveland Clinic dataset (302 observations). The Cross-Validation standard deviation of 0.07 indicates that the model experiences some variance depending on the patient sample, which is typical for datasets under 1,000 records.

Future Improvements: For future iterations, I will explore ensemble methods like Random Forest or Gradient Boosting to see if combining multiple decision trees can push the validated accuracy closer to 90%.

In []: